Abstract

Triple-negative breast cancer, or TNBC, is a particularly difficult form of breast cancer because it does not have the estrogen receptor, progesterone receptor and HER2 proteins.” In the present study, utilized intelligent computational tools of biological data integrated with machine learning to identify potential markers for diagnosis of TNBC and novel therapeutic targets.

Gene information was obtained from a publicly available dataset of TNBC patients and healthy breast tissue (GSE38959). Then used a tool called limma to examine how the genes re expressed differently the TNBC and healthy samples. With this gene that re significantly more or less active than expected, based on some statistical cutoffs. plotted this info with things like volcano plots, principal component analysis and heatmaps and it was clear that there was a big difference been the TNBC and normal samples. To pinpoint the most important marker genes, used some machine learning tricks, like Random Forest and LASSO regression.

This helped us find central 'hub' genes, like CDC2 and HSP90AA1, which play a big role in controlling cell division and how cancer grows. By checking how drugs interact with these genes, found some promising candidates that might be good targets for new treatments. even used a method called SHAP to understand exactly how each gene contributed to our prediction model.

So, looking at everything, this research pointed out several important genes — including KLHL13, CORIN, PIK3C2G, KLK14, STMN1, CDC2, and HSP90AA1. These could potentially be used as markers to help diagnose TNBC or as targets for new treatments. The way combined different methods in this study offers a good starting point for really understanding how TNBC works at a molecular level. It also sets the stage for doctors and researchers to test these findings further in the clinic and lab.

Keywords:

Triple-negative breast cancer (TNBC), Gene expression profiling, Bioinformatics, Differentially expressed genes (DEGs), Gene Ontology (GO), KEGG pathways, Gene Set Enrichment Analysis (GSEA), Machine learning, Random Forest, LASSO regression, ROC analysis, Protein–protein interaction (PPI), Hub genes, Drug–gene interaction, SHAP analysis

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Abbreviations Table

Abbreviation	Full Form
TNBC	Triple-Negative Breast Cancer
GEO	Gene Expression Omnibus
DEG	Differentially Expressed Genes
PCA	Principal Component Analysis
GO	Gene Ontology
KEGG	Kyoto Encyclopedia of Genes and Genomes
GSEA	Gene Set Enrichment Analysis
ROC	Receiver Operating Characteristic
AUC	Area Under the Curve
PPI	Protein–Protein Interaction
STRING	Search Tool for the Retrieval of Interacting Genes/Proteins
ML	Machine Learning
RF	Random Forest
LASSO	Least Absolute Shrinkage and Selection Operator
SHAP	SHapley Additive exPlanations
TCGA	The Cancer Genome Atlas

ER	Estrogen Receptor
PR	Progesterone Receptor
HER2	Human Epidermal Growth Factor Receptor 2
DNA	Deoxyribonucleic Acid
RNA	Ribonucleic Acid
mRNA	Messenger RNA
qRT-PCR	Quantitative Real-Time Polymerase Chain Reaction
OS	Overall Survival
HR	Hazard Ratio
CI	Confidence Interval

Introduction

Triple-negative BRCAs are aggressive subtypes of Breast cancer is one of the most common malignancies worldwide and a cause of death in women. It is a heterogeneous disease, comprising several different molecular subtypes with varying clinical and biological characteristics. characterized by lacking estrogen receptor, progesterone receptor and human epithermal growth factor receptor 2 (HER2). In these patients, primary chemotherapy is useful however it is unaffected by ovarian suppression, radiotherapy or targeted therapy against all subtypes. Therefore, further insight into the disease pathophysiology is required for better treatments. Recently, microarrays technology has allowed for the investigation of the expression of thousands of genes at one time. The GSE38959 human transcriptome dataset will be used in this project. It is known that such large data provides a source of biological significant information but extracting such signal is a challenge.

The following bioinformatics methods will be utilised; analysis of differentially expressed genes (DEGs), pathway and protein interaction networks, functional enrichment (using gene ontology and KEGG pathway enrichment) and gene set enrichment analysis (GSEA) will be performed. Machine learning methods (LASSO and random forest) and ROC analysis will provide further insight. PPI network analysis was also used to identify hub genes that important for disease pathogenesis. Drug-gene interaction analysis could be used to obtain potential drugs or therapeutic targets. Explainable AI (SHAP) could be used to uncover gene contribution to prediction. These tools used in combination would give a detailed look into TNBC at the molecular level.

This work would seek to combine bioinformatics and machine learning methods to search for potential biomarkers and therapeutic targets related to TNBC. Integrating differential expression and functional annotation analysis, network-based approaches, and precise feature selection as well as prediction models, this work offer a full picture in understanding the molecular mechanisms of TNBC.

Review of Literature

Triple-negative breast cancer (TNBC) is tough—plain and simple. Out of all the breast cancer types, it's one of the hardest to treat. The problem? TNBC doesn't have the three big receptors—estrogen, progesterone, and HER2—that many treatments target. So, doctors don't have those go-to drugs that work for most other cases. It's aggressive, spreads quickly, and the outlook is often worse than what you see with other breast cancers (Carey et al., 2010; Bauer et al., 2007).

You'll find TNBC in somewhere between 15% and 20% of breast cancer patients, and it tends to hit younger women more often. These tumours usually show up higher-grade, and there's always this nagging risk that the cancer will come back, even after treatment ends (Fitzmaurice et al., 2018; Hwang et al., 2019). Right now, chemotherapy is still the main weapon doctors reach for. It's something, but the cancer can become resistant, and relapses happen way too often (Hwang et al., 2019).

Xiao and their team (2021) really hammered this home. Because TNBC cells don't have the usual targets, treatment basically comes down to chemo, surgery, and sometimes radiation. Even then, there's a serious chance the cancer shows up somewhere else later, mostly because no two tumours are exactly the same and resistance builds over time. The study also points out that several signalling pathways—like Notch, Wnt/β-catenin, Hedgehog, and TGF-β—drive TNBC's growth and ability to spread (Hwang et al., 2019).

This is where technology is beginning to make a difference. Access to innovative techniques such as microarray and transcriptomics, enables researchers to look inside the cell and identify the genes that are being affected (Barrett et al, 2013; Komatsu et al, 2013).

Open data sites, like the "Gene Expression Omnibus (GEO)", have made it a lot simpler to find new biomarkers and then check if they actually work.

When scientists look at genes using tools like GO and KEGG, they keep finding the same thing: TNBC tumours are full of genes that push cells to divide, copy their DNA, and move through their life cycle. It's no surprise these cancers grow so quickly and are really tough to stop (Huang et al., 2009; Hanahan and Weinberg, 2011)

Researchers have started to lean on machine learning to make sense of all this data. Techniques like Random Forest and LASSO regression help spotlight the genes that really move the needle in diagnosis and prognosis. ROC analysis comes in to test just how well these genes predict the disease (Zhao et al., 2018; Luo et al., 2019).

Finding the drivers doesn't stop at just genes, either. With protein-protein interaction networks, researchers use platforms like STRING and Cytoscape to map out how important genes—like ASPM, CENPK, MELK, and NEK2—work together and fuel tumour growth (Li et al., 2018; Lv et al., 2019; Szklarczyk et al., 2019; Shannon et al., 2003).

And while new treatments are always on the horizon—things like PARP inhibitors, EGFR inhibitors, and mTOR inhibitors—these, too, run into the wall of resistance and the wild diversity of TNBC tumours. There's promise in immunotherapy, nanomedicine, and artificial intelligence shaking things up, but we're not there yet. We've got a long road ahead before these new strategies truly tip the balance for patients (Hwang et al., 2019).

Material and Methods

Data Acquisition

From the GEO database - home to large-scale gene activity records - came information on triple-negative breast cancer. That collection, known as GSE38959, appeared first in work led by Komatsu and team back in 2013. Sitting inside it genetic readouts from three dozen tumour tissues alongside thirteen healthy cell samples taken from milk ducts. Scientists built these profiles using tiny chips designed to track thousands of genes at once. (Xiao X,2021)

Data Preprocessing and Normalization

Data handling started with the R software for processing raw signals. Using reference files, probe identifiers got linked to known gene labels instead. Those probes missing correct matches dropped out early on. When several probes pointed to one gene, extras vanished so just a single label stayed behind. Only samples labelled TNBC or healthy made it into later steps somehow. Good cleanup at first keeps results sharp when spotting active gene shifts (Kulasingam and Diamandis, 2008).

Differential Expression Gene Analysis

To compare gene activity in triple-negative breast cancer versus healthy tissue, researchers applied the limma tool in R. Fitting a straight-line pattern to the measurements came first, then used empirical Bayes techniques to sharpen result reliability. Only those genes showing a corrected significance level below alongside a doubling or halving of activity stood out as meaningfully changed (Xiao X,2021).

Gene Expression Patterns Shown Visually

Out of R tools like ggplot2 and pheatmap came volcano plots, PCA views, and heatmaps - each built to show how gene activity shifts. Clustering trends pop up clearly when TNBC tissue sits beside healthy cells in these displays. Expression swings stand out, shaped by methods pulled from past work (Li et al., 2018). Visuals unfold piece by piece, guided by code that maps divergence without smoothing edges.

Gene Function and Pathway Mapping

Looking at gene patterns started with tools inside an R software package called cluster Profiler, based on work by (Yu 2012). Instead of guessing, this method spots biological jobs, tiny cell activities, or chemical paths that show up more than expected in shifted genes. Because of these shifts, meaning behind altered genes becomes clearer - a point shown earlier by Huang's research in 2009.

Gene Set Enrichment Analysis

Looking at patterns in gene activity, a method called Gene Set Enrichment Analysis helped spot which known groups of genes stood out across ordered results. These groups came from the Molecular Signatures Database, detailed by (Liberzon 2015). Instead of just using strict cutoffs for active genes, this approach reveals shifts in biological pathways more smoothly.

Using Machine Learning to Find Biological Markers

Out of large sets of genes, a few stand out when tested through pattern-finding tools. One method builds many decision trees to spot consistent trends across data splits. Another pulls forward only the strongest contributors by shrinking weak signals toward zero. Both have shown up often in DNA-based research looking for reliable markers. Their track record includes sorting meaningful patterns from background noise in complex datasets.

Receiver Operating Characteristic Analysis

A graph showing how well certain genes can tell apart conditions came from an R tool called pROC. This plot's fullness, known as AUC, gave a measure of how strong each marker is at pointing to outcomes (Zhao et al., 2018).

Building networks of interacting proteins

From the STRING database, a web of protein connections came together, pulling in both confirmed and forecasted links between proteins (Szklarczyk et al., 2019). With graph methods guiding the view, the layout took shape, revealing central genes through their ties.

Hub Gene Identification

Connections across the network helped spot key genes, thanks to how many links they carried (Freeman, 1978). Top-ranked ones drew attention because their reach suggested strong influence over TNBC development.

Drug–Gene Interaction Analysis

Looking at how drugs interact with genes through public databases opened paths for possible treatments. With this method, scientists can spot medicines likely to act on key genes. Clues about ways to treat diseases start to appear from such analysis (Guo et al., 2017).

SHAP-Based Model Interpretation

One way to see how a model makes choices is through SHAP. Each gene's role in shaping predictions becomes clearer when viewed through its SHAP value. Instead of guessing, we observe what shifts outcomes most. Behind every decision lies a mix of influences - one stands out more than others at times. This method comes from work by (Lundberg Lee 2017).

Results:

Table 1: Differential expression gene table

Genes	AveExpr	t	P.Value	adj.P.Val	Base mean
KLHL13	207.3381	-12.1115	3.15E-15	6.22E-11	1.649084
HOXA4	457.3349	-10.281	5.42E-13	5.10E-09	1.062731
PAR5	36.99581	-10.1593	7.75E-13	5.10E-09	1.017157
DST	1145.728	-9.57884	4.38E-12	2.16E-08	0.786723
ANKRD30A	463.954	-9.47036	6.08E-12	2.40E-08	0.74118
DMD	632.3549	-9.20331	1.37E-11	4.52E-08	0.625575
ABCA5	649.8605	-8.96908	2.82E-11	7.96E-08	0.51998
TSHZ2	514.6621	-8.80286	4.73E-11	1.03E-07	0.442588
NTN4	899.8598	-8.79065	4.91E-11	1.03E-07	0.436823
FLJ11292	1202.442	-8.75574	5.48E-11	1.03E-07	0.420275
STC2	264.0119	-8.74054	5.74E-11	1.03E-07	0.413041
PIK3C2G	90.29651	-8.69125	6.70E-11	1.10E-07	0.38946
CCL28	287.1647	-8.63977	7.87E-11	1.18E-07	0.364633
CBX3	7784.884	8.620091	8.37E-11	1.18E-07	0.355089
EDN3	134.2242	-8.57133	9.75E-11	1.26E-07	0.331311
LOC728264	1270.443	-8.55619	1.02E-10	1.26E-07	0.323893
C4orf18	206.5733	-8.47107	1.34E-10	1.55E-07	0.281841
PDGFA	2851.953	-8.40183	1.66E-10	1.82E-07	0.247222
KLK14	211.1442	-8.28237	2.43E-10	2.52E-07	0.186594
PI15	390.6912	-8.12476	4.00E-10	3.83E-07	0.104866
CBX7	2521.116	-8.11939	4.07E-10	3.83E-07	0.102042
CORIN	38.24884	-7.94976	7.00E-10	6.29E-07	0.011758
ERBB4	100.1779	-7.89661	8.31E-10	7.13E-07	-0.01701
LOC100128979	597.8	-7.85288	9.56E-10	7.86E-07	-0.04086
ANLN	87.74605	7.717945	1.48E-09	1.13E-06	-0.11545
GRAMD3	209.0744	-7.71432	1.49E-09	1.13E-06	-0.11747
HEMK1	57.57209	-7.69326	1.60E-09	1.17E-06	-0.12926
LOC442245	131.7042	-7.67062	1.72E-09	1.21E-06	-0.14197
GSTM2	312.8526	-7.5967	2.18E-09	1.49E-06	-0.18376
TUBA1B	43986.05	7.543984	2.59E-09	1.70E-06	-0.21385
NUSAP1	480.4902	7.508152	2.91E-09	1.82E-06	-0.23443
GPRASP1	136.77	-7.49472	3.04E-09	1.82E-06	-0.24217
CX3CR1	337.6372	-7.49405	3.04E-09	1.82E-06	-0.24256
AK5	666.2295	-7.4821	3.16E-09	1.84E-06	-0.24947
TUBA1C	917.0465	7.459194	3.41E-09	1.92E-06	-0.26273
ARHGAP19	109.8767	-7.43624	3.67E-09	2.01E-06	-0.27607
HIST1H2BE	510.7884	7.41227	3.97E-09	2.10E-06	-0.29004
C5orf26	947.1163	-7.4068	4.04E-09	2.10E-06	-0.29324
DKK3	360.1977	-7.34081	5.01E-09	2.52E-06	-0.33199

Table 2: Differential expression gene table showing upregulated genes

Genes	Ave Expr	t	P.Value	adj.P. Val
CBX3	7784.884	8.620091	8.37E-11	1.18E-07
ANLN	87.74605	7.717945	1.48E-09	1.13E-06
TUBA1B	43986.05	7.543984	2.59E-09	1.70E-06
NUSAP1	480.4902	7.508152	2.91E-09	1.82E-06
TUBA1C	917.0465	7.459194	3.41E-09	1.92E-06
HIST1H2BE	510.7884	7.41227	3.97E-09	2.10E-06
ASF1B	71.90698	7.153876	9.21E-09	4.04E-06
UBE2C	2298.698	7.121823	1.02E-08	4.30E-06
HIST2H2BE	2252.186	6.837775	2.59E-08	8.21E-06
HIST1H2BG	1290.928	6.781505	3.12E-08	9.06E-06
E2F1	80.9393	6.735122	3.63E-08	9.97E-06
HIST1H3B	198.7767	6.682471	4.32E-08	1.14E-05
SNRPG	9710.93	6.580296	6.05E-08	1.53E-05
PSMD12	333.093	6.566058	6.34E-08	1.57E-05
HIST1H2BD	872.4651	6.56174	6.43E-08	1.57E-05
HIST1H2BL	1029.965	6.558916	6.49E-08	1.57E-05
HIST1H2AH	410.9302	6.556318	6.55E-08	1.57E-05
CCNE1	123.6098	6.521989	7.33E-08	1.70E-05
MLF1IP	98.26674	6.508693	7.66E-08	1.76E-05
GPR172A	121.414	6.48214	8.36E-08	1.84E-05
TMSB10	1409.093	6.479721	8.42E-08	1.84E-05
TPX2	342.654	6.442953	9.51E-08	2.00E-05
UBE2T	1229.249	6.424109	1.01E-07	2.10E-05
HIST1H2BF	1213.512	6.385017	1.15E-07	2.30E-05
CCNB1	333.5698	6.362284	1.24E-07	2.43E-05
H2AFV	3517.372	6.34107	1.33E-07	2.55E-05
HIST1H2BO	642.6	6.334546	1.36E-07	2.55E-05
TTK	91.02698	6.329777	1.38E-07	2.55E-05
TPM3	1018.998	6.323629	1.41E-07	2.58E-05
KPNA2	6290.698	6.291807	1.57E-07	2.77E-05
HIST3H2A	3230.14	6.290603	1.57E-07	2.77E-05
HIST1H2BC	754.8744	6.257091	1.76E-07	3.04E-05
GGCT	18223.02	6.235378	1.89E-07	3.21E-05
CKS2	2009.163	6.216567	2.01E-07	3.39E-05
SF3A3	2023.349	6.199869	2.12E-07	3.52E-05
LOC100130289	16573.72	6.18438	2.23E-07	3.64E-05
HIST1H2BI	628.1326	6.179517	2.27E-07	3.64E-05
OIP5	307.4767	6.15433	2.46E-07	3.93E-05
RACGAP1	308.5581	6.13751	2.61E-07	4.04E-05
CDCA5	186.027	6.135849	2.62E-07	4.04E-05
DEPDC1	176.1302	6.126311	2.70E-07	4.14E-05
TUBB2A	130.2116	6.104497	2.90E-07	4.35E-05
H2AFZ	1862.233	6.099764	2.95E-07	4.37E-05

Table 3: Differential expression gene table showing downregulated genes

Genes	Ave Expr	t	P.Value	adj.P. Val
KLHL13	207.3381	-12.1115	3.15E-15	6.22E-11
HOXA4	457.3349	-10.281	5.42E-13	5.10E-09
PAR5	36.99581	-10.1593	7.75E-13	5.10E-09
DST	1145.728	-9.57884	4.38E-12	2.16E-08
ANKRD30A	463.954	-9.47036	6.08E-12	2.40E-08
DMD	632.3549	-9.20331	1.37E-11	4.52E-08
ABCA5	649.8605	-8.96908	2.82E-11	7.96E-08
TSHZ2	514.6621	-8.80286	4.73E-11	1.03E-07
NTN4	899.8598	-8.79065	4.91E-11	1.03E-07
FLJ11292	1202.442	-8.75574	5.48E-11	1.03E-07
STC2	264.0119	-8.74054	5.74E-11	1.03E-07
PIK3C2G	90.29651	-8.69125	6.70E-11	1.10E-07
CCL28	287.1647	-8.63977	7.87E-11	1.18E-07
EDN3	134.2242	-8.57133	9.75E-11	1.26E-07
LOC728264	1270.443	-8.55619	1.02E-10	1.26E-07
C4orf18	206.5733	-8.47107	1.34E-10	1.55E-07
PDGFA	2851.953	-8.40183	1.66E-10	1.82E-07
KLK14	211.1442	-8.28237	2.43E-10	2.52E-07
PI15	390.6912	-8.12476	4.00E-10	3.83E-07
CBX7	2521.116	-8.11939	4.07E-10	3.83E-07
CORIN	38.24884	-7.94976	7.00E-10	6.29E-07
ERBB4	100.1779	-7.89661	8.31E-10	7.13E-07
LOC100128979	597.8	-7.85288	9.56E-10	7.86E-07
GRAMD3	209.0744	-7.71432	1.49E-09	1.13E-06
HEMK1	57.57209	-7.69326	1.60E-09	1.17E-06
LOC442245	131.7042	-7.67062	1.72E-09	1.21E-06
GSTM2	312.8526	-7.5967	2.18E-09	1.49E-06
GPRASP1	136.77	-7.49472	3.04E-09	1.82E-06
CX3CR1	337.6372	-7.49405	3.04E-09	1.82E-06
AK5	666.2295	-7.4821	3.16E-09	1.84E-06
ARHGAP19	109.8767	-7.43624	3.67E-09	2.01E-06
C5orf26	947.1163	-7.4068	4.04E-09	2.10E-06
DKK3	360.1977	-7.34081	5.01E-09	2.52E-06
CMYA5	46.34256	-7.33458	5.11E-09	2.52E-06
FAM47C	146.786	-7.31165	5.50E-09	2.62E-06
FAM47E	222.2698	-7.30746	5.58E-09	2.62E-06
ZBTB4	2509.116	-7.19741	7.99E-09	3.67E-06
TP63	260.6384	-7.18956	8.20E-09	3.68E-06
SPARCL1	193.9547	-7.12323	1.02E-08	4.30E-06
CYP1A2	920.9302	-7.10669	1.07E-08	4.42E-06
FAM13B1	171.0628	-7.0817	1.17E-08	4.70E-06
RAI2	228.8463	-7.05648	1.27E-08	5.00E-06
OR4D2	156.9977	-6.98166	1.62E-08	6.26E-06

Differential Expression and Global Transcriptomic Patterns

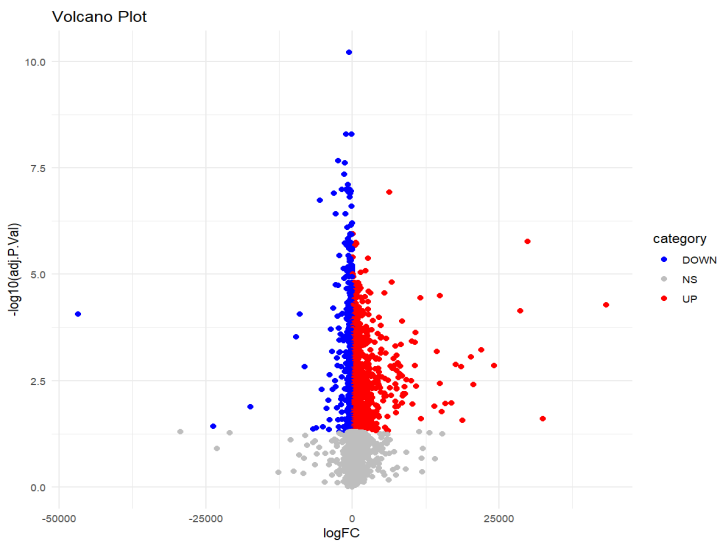

Fig:1 Volcano plot showing gene expression

Strong transcriptional reprogramming in TNBC was indicated by the volcano plot, which showed a distinct split between upregulated and downregulated genes. While a specific selection of genes was considerably downregulated, indicating inhibition of normal cellular regulatory systems, a large number of genes shown considerable upregulation (logFC > 1, adjusted p-value < 0.05), indicating activation of carcinogenic pathways.

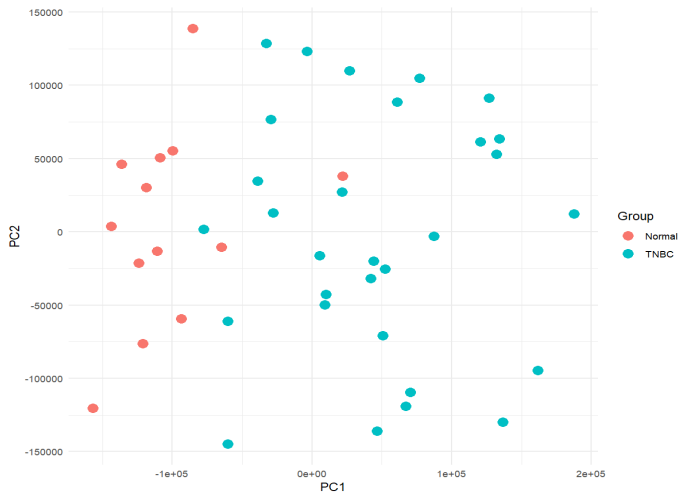

Fig:2 PCA plot showing gene expression

PCA clearly divided the TNBC and normal samples into two distinct groups, demonstrating the strength of our findings. This distinct split validates the biological significance of our data and implies that there were few technical errors influencing the samples. The variations we

saw also suggest that the gene expression alterations we discovered may be useful for diagnosis as they are sufficient to distinguish a tumour from healthy tissue.

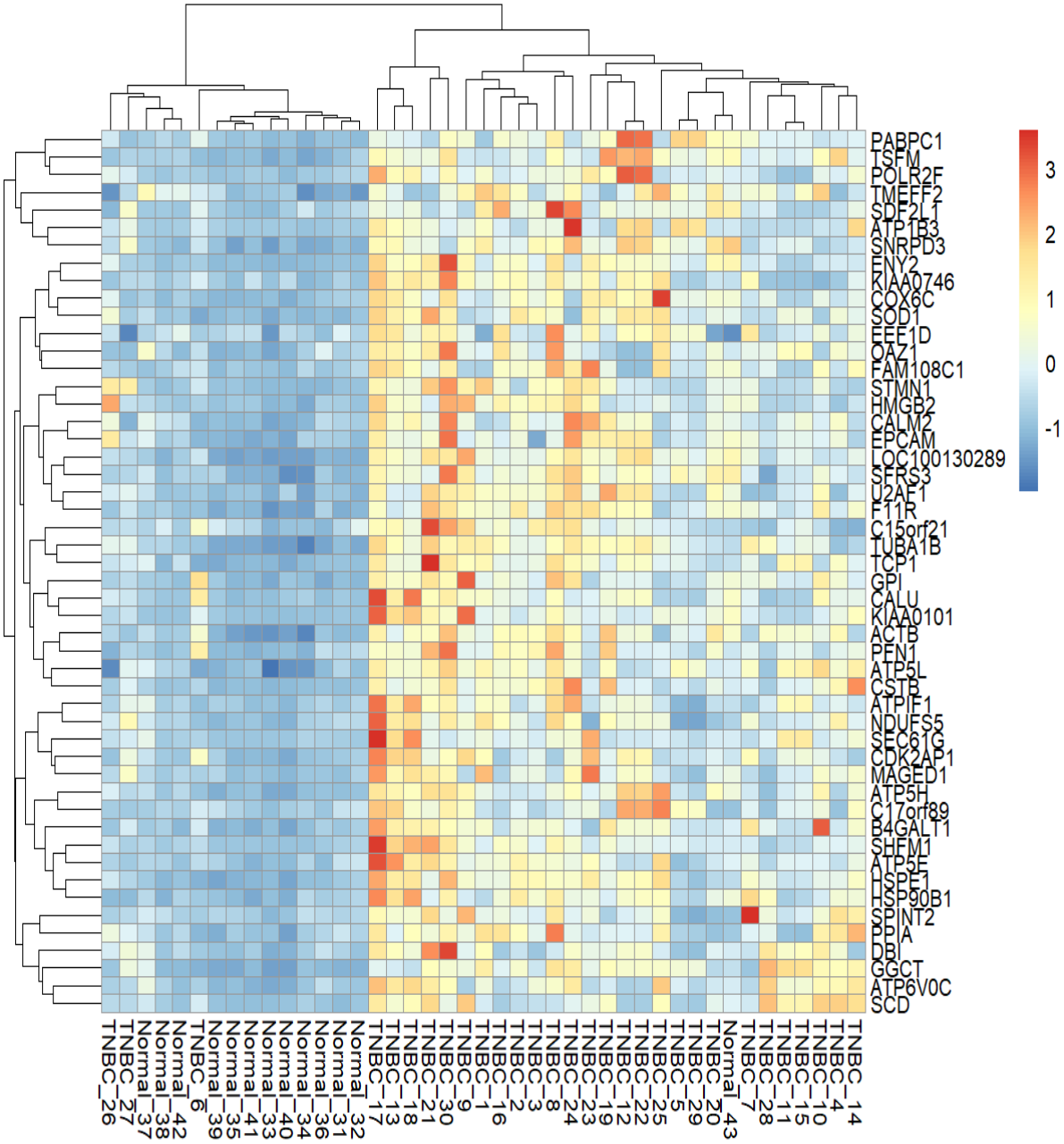

Fig: 3 Heatmap showing top 50 differential expression genes

The heatmap of the top 50 upregulated genes showed very strong and stable overexpression patterns of these genes in TNBC samples compared to normal controls. These genes most probably constitute major players in the development of tumours and could be a source of candidate biomarkers or therapeutic targets.

Functional Enrichment and Pathway Analysis

GO functional annotation analysis, the DEGs were found to be associated with biological functions involved in cell cycle progress, mitosis, chromosomal segregation, and DNA replication. These cellular activities are characteristic of fast-growing malignant tumour cells and correlate well with the highly aggressive nature of triple-negative breast cancer.

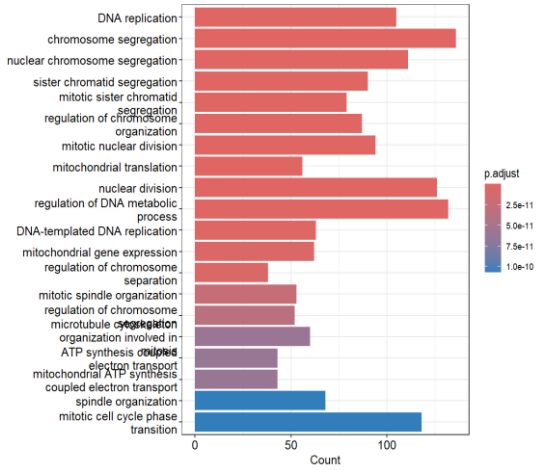

Fig:4 Go enrichment bar plot

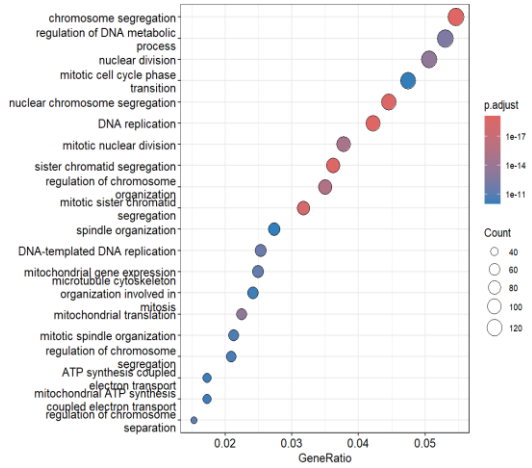

Fig: 4.1 Go enrichment dot plot

KEGG pathway analysis confirmed these observations by detecting an over-representation of pathways including cell cycle regulation, DNA replication, p53 signalling, and cancer-related pathways. Aberrant p53 pathway is the most interesting among them as it is of paramount importance for genomic stability and apoptosis and its loss is very often connected with tumour development and resistance to treatment in TNBC.

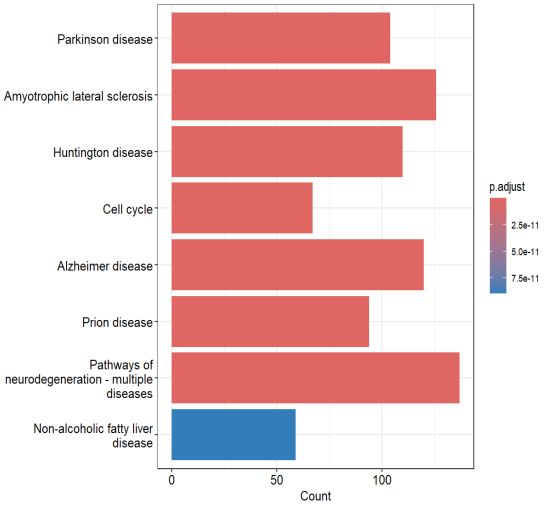

Fig: 5 KEGG enrichment bar plot

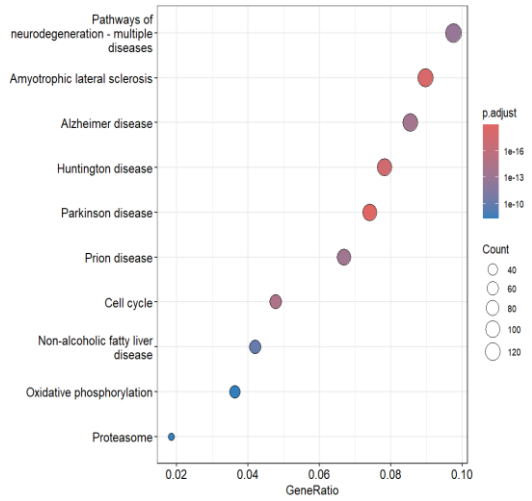

Fig: 5.1 KEGG enrichment dot plot

Gene Set Enrichment Analysis (GSEA) provided further information by examining the gene expression profiles within the complete transcriptome without imposing any cutoffs. The analysis revealed an enrichment for the proliferation and oncogenesis pathway within the TNBC patients, implying that specific molecular processes were involved in promoting tumour growth.

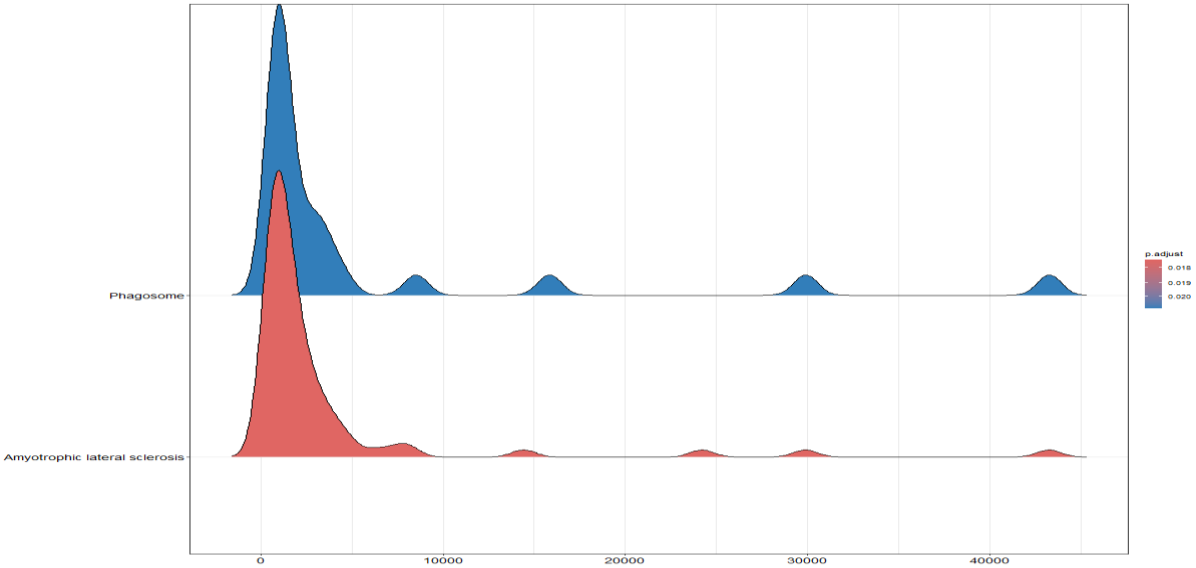

Fig:6 GSEA_ridgeplot

Protein–Protein Interaction Network and Hub Gene Analysis

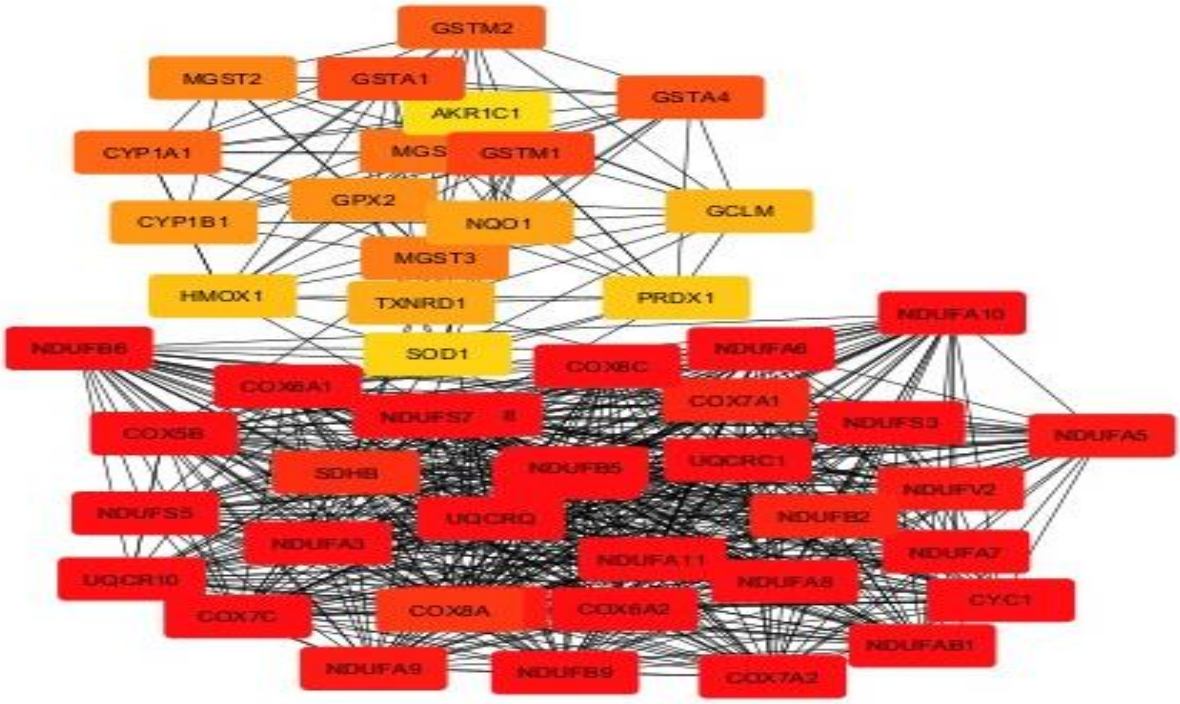

Fig:7 Protein–Protein Interaction Network and Hub Gene Analysis

The protein protein interaction (PPI) network derived from these genes presents a tightly woven network primarily centred on mitochondrial function and the regulation of oxidative stress in triple-negative breast cancer (TNBC). Genes such as CYC1, COX families (COX5B COX6A1 COX6C, COX7A1 COX7A2 COX7C, COX8A), and several NDUF subunits (NDUFA NDUFB NDUFS families) cluster together showing a pronounced overrepresentation of electron transport chain (ETC) and oxidative phosphorylation components. It points out that TNBC tumour cells may undergo metabolic reprogramming and become highly dependent on mitochondrial function for energy production and survival. The involvement of SDHB and UQCR proteins adds weight to mitochondrial respiration and energy production. At the same time, a second large functional group is made up of genes responsible for detoxification and antioxidant defense such as GSTM1 GSTA1 GSTA4, MGST1-3 GPX2 SOD1, PRDX1 TXNRD1 NQO1, HMOX1, and GCLM. They play very important roles in regulating reactive oxygen species (ROS) and cell redox balance, which means that TNBC cells undergo stress induced operations by oxygen to ensure cell death prevention and promotion of their growth. Besides CYP1A1 CYP1B1, and AKR1C1 genes are highly associated with xenobiotic metabolism and chemotherapy drug resistance. The network overall delineates two major biological themes: Power production from mitochondria and controlling oxidation stress response and detoxification mechanisms. Oxidative stress response and detoxification systems

The results indicate that apart from uncontrollable cell proliferation, TNBC development occurs due to metabolic changes and increased oxidative resistance. This makes cancerous cells resistant to stress. It means that manipulating mitochondria metabolism and antioxidant defense systems could be an effective treatment strategy for TNBC patients.

Drug–Gene Interaction and Therapeutic Implications

The drug-gene interaction analysis indicated some drugs that could be used to treat the hub genes. This means that there are various genes involved in drug interactions that can be used to repurpose existing drugs.

In view of the lack of targeted drugs for TNBC, this discovery has provided critical insights into potential therapies. Experimental verification is necessary to validate the effectiveness of these drug-gene interactions.

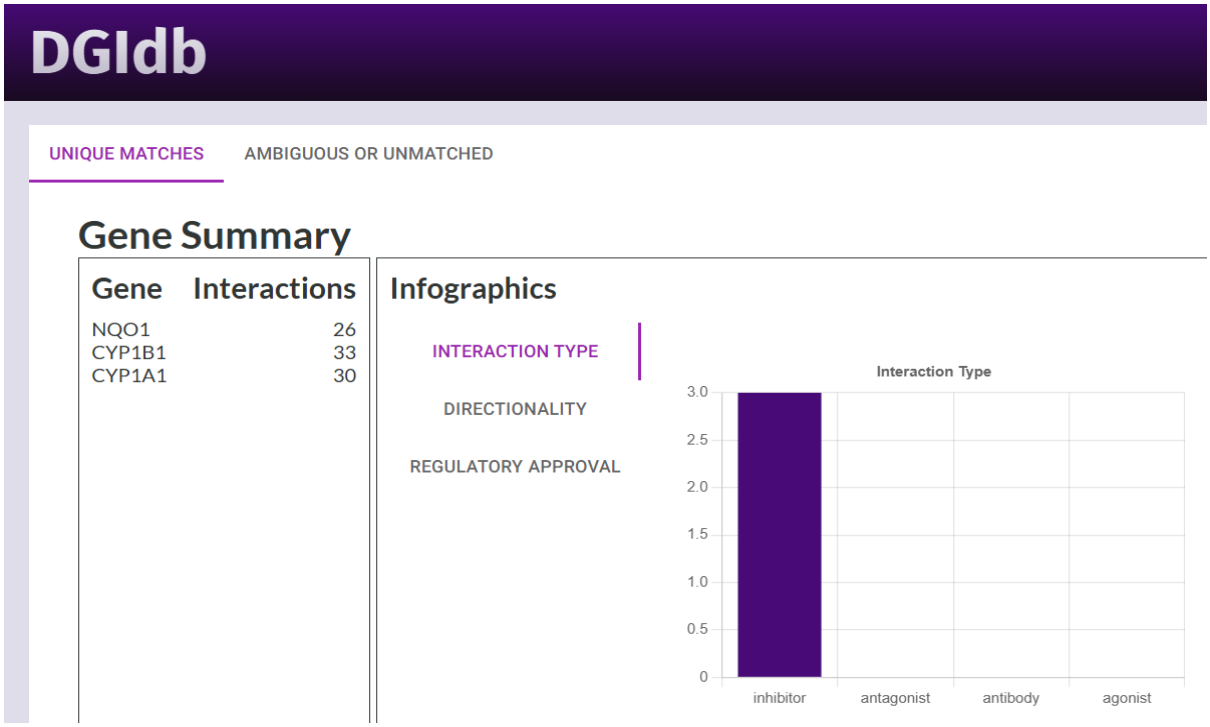

Fig: 8 Gene interaction types using DGIdb database

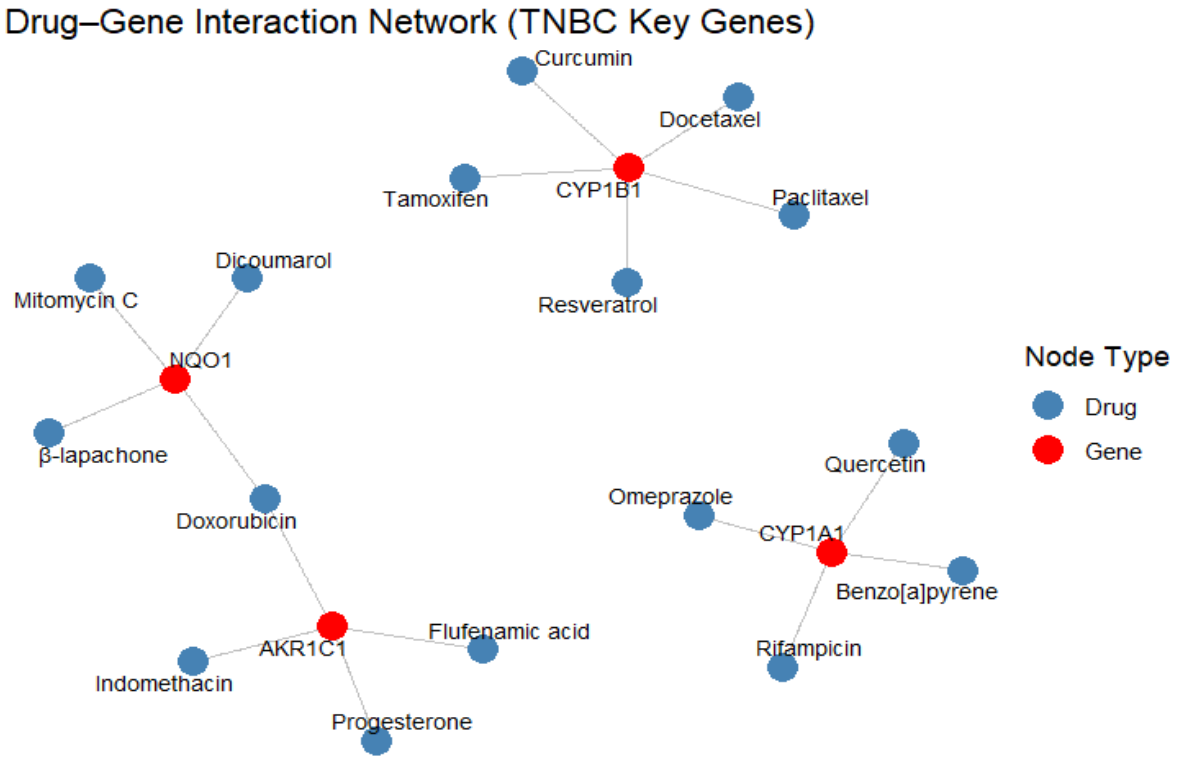

Fig: 9 Drug–gene interaction network

Table 4: Drug-gene interaction table

Gene	Drug	Indication
NQO1	DICUMAROL	Anticoagulants
NQO1	CYTARABINE	antineoplastic agent
NQO1	ACETAMINOPHEN	analgesic
NQO1	DOXORUBICIN HYDROCHLORIDE	antineoplastic agent
NQO1	ETOPOSIDE	Antineoplastic Agents
NQO1	METHOTREXATE	DMARD
NQO1	ETHACRYNIC ACID	Antihypertensive Agents; Diuretics
NQO1	IDARUBICIN	Antineoplastic Agents
CYP1A1	VALPROIC ACID	Anticonvulsants, for treatment of basal cell carcinoma, anticonvulsant
CYP1A1	LIOTHYRONINE	Hormone Replacement Agents
CYP1A1	PHENYTOIN SODIUM	Anticonvulsants
CYP1A1	GRANISETRON	antiemetic
CYP1A1	PHENOBARBITAL	Anticonvulsants; Hypnotics and Sedatives
CYP1A1	NILOTINIB	antineoplastic agent
CYP1A1	DACARBAZINE	Antineoplastic Agents
CYP1A1	PROMETHAZINE HYDROCHLORIDE	Hypnotics and Sedatives; Anti-anxiety agents; Anti-allergic Agents
CYP1A1	RISPERIDONE	Antipsychotic Agents, antipsychotic agent
CYP1A1	CARBAMAZEPINE	for treatment of bipolar disorder
CYP1A1	CLOZAPINE	Antipsychotic Agents
CYP1A1	DAUNORUBICIN LIPOSOMAL	antineoplastic agent
CYP1A1	DOXORUBICIN HYDROCHLORIDE	antineoplastic agent
CYP1A1	OLANZAPINE	for treatment of bipolar disorder,antidepressant,antipsychotic agent
CYP1A1	SPIRONOLACTONE	antihypertensive agent
CYP1A1	DOCETAXEL ANHYDROUS	antineoplastic agent
CYP1B1	CODEINE ANHYDROUS	Opioid; Antitussive Agents, Analgesics
CYP1B1	RISPERIDONE	Antipsychotic Agents, antipsychotic agent
CYP1B1	LETROZOLE	antineoplastic agent
CYP1B1	TOREMIFENE	SERM,antineoplastic agent
CYP1B1	TRAMADOL	analgesic
CYP1B1	AMINOGLUTETHIMIDE	Antineoplastic agents
CYP1B1	EXEMESTANE	antineoplastic agent
CYP1B1	DOXORUBICIN HYDROCHLORIDE	antineoplastic agent
CYP1B1	SPIRONOLACTONE	antihypertensive agent
CYP1B1	ANASTROZOLE	antineoplastic agent

Biomarker Identification and Predictive Modelling

Table 5: Top 5 evaluated genes table

Gene	AUC	Function
KLHL13	0.994872	Serine protease that acts on cancer cell proliferation through the degradation of insulin-like growth factor-binding proteins (IGFBP) and PAR-2 activation. This enzyme is often investigated along with other KLKs such as KLK2, KLK5, and KLK13 in relation to cancer development especially in prostate and colon cancer.
CORIN	0.987179	Mutations in the CORIN gene are associated with hypertension and heart diseases, including left ventricular hypertrophy
PIK3C2G	0.982051	Related to the PI3K/AKT/mTOR pathway, which is critical for cell proliferation and survival. While PIK3CA is the primary oncogene in this family, other PI3K members are often studied in the context of cancer growth regulation.
KLK14	0.982051	Identified as a potential prognostic and/or predictive biomarker in Ovarian Cancer and Colon Cancer as prolonged progression-free survival (PFS) and overall survival (OS), acting as a potential tumour suppressor
PAR5	0.979487	PARs are involved in cancer-related signalling pathways, often regulated by kallikrein proteins.

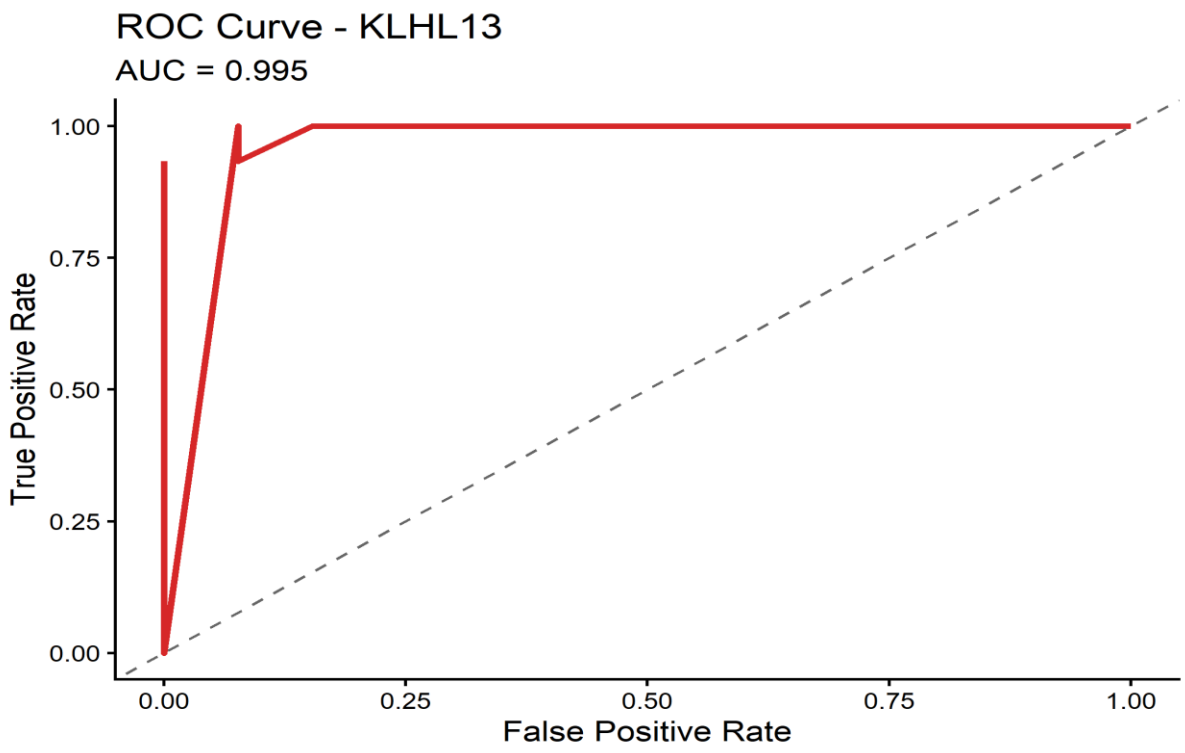

Fig:10 ROC curve of top gene

To evaluate in order to assess the diagnostic capability of the selected genes, ROC analysis was conducted. A number of genes exhibited excellent AUC scores, which is a clear indication of a high level of discrimination between TNBC tissues and normal tissues. The most promising gene showed almost perfect discriminatory capability, which can be considered as evidence of its role as a robust biomarker. The collective ROC curves of the selected candidate genes showed consistent discriminatory power as well. This result confirms earlier findings that suggest the significance of multi-gene biomarkers over single biomarkers.

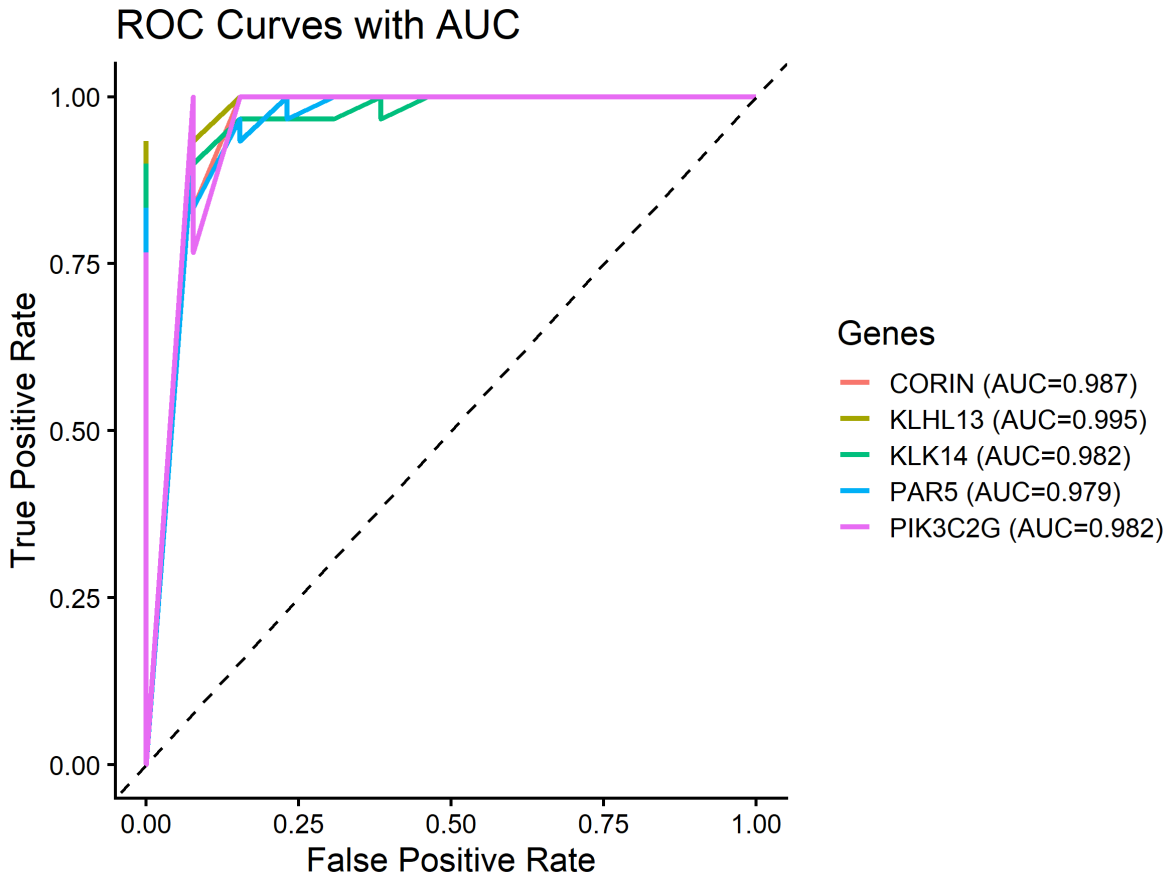

Fig:10.1 ROC curve of top 5 genes

Machine learning models like Random Forest and LASSO regression algorithms were applied in order to select relevant features and validate the models. The Random Forest classifier recognized highly important genes that contributed substantially to the classification process. On the other hand, LASSO regression was successfully used to reduce the dimensionality of the data set and select a number of genes with the most significant predictive power.

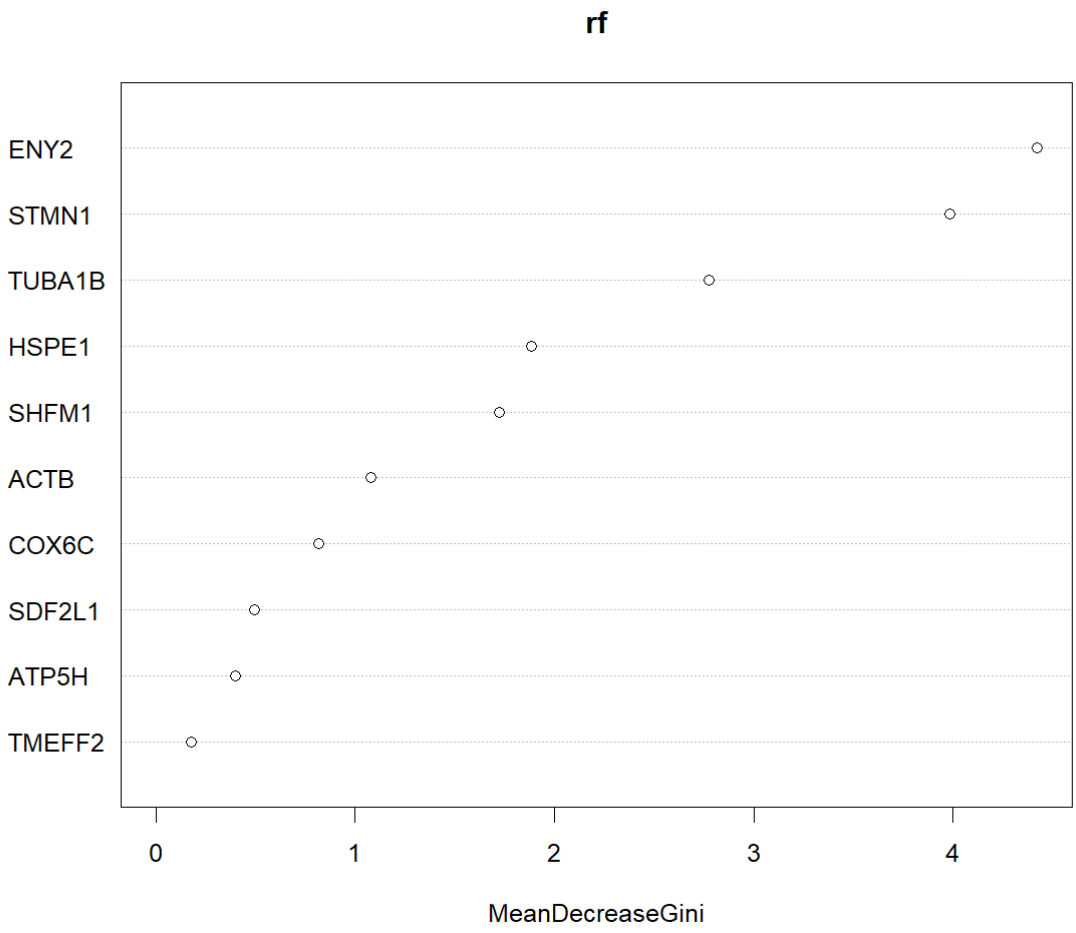

Fig:11 Random Forest model key genes

Table 6: Random forest model Top evaluated genes table

Gene	**AUC**	**Importance**
ACTB	0.930769	Often upregulated in aggressive tumours; used as a housekeeping gene but not functionally inert in cancer
TMEFF2	0.776923	Frequently downregulated in aggressive cancers, potential biomarker of loss of suppression
TUBA1B	0.974359	Strongly linked to high proliferative index and chemotherapy sensitivity
HSPE1	0.95641	Supports metabolic and stress adaptation in fast-growing tumours
COX6C	0.923077	Cancer cells often rewire oxidative phosphorylation; COX complex genes can be upregulated
STMN1	0.969231	strong proliferation and poor prognosis marker, linked to chemotherapy response
SHFM1	0.960256	Indirect role via protein degradation pathways
ATP5H	0.914103	Supports high metabolic demand in aggressive tumours
SDF2L1	0.864103	Helps tumour cells survive hypoxia and stress conditions
ENY2	0.935897	Supports transcriptional reprogramming in rapidly dividing cancer cells

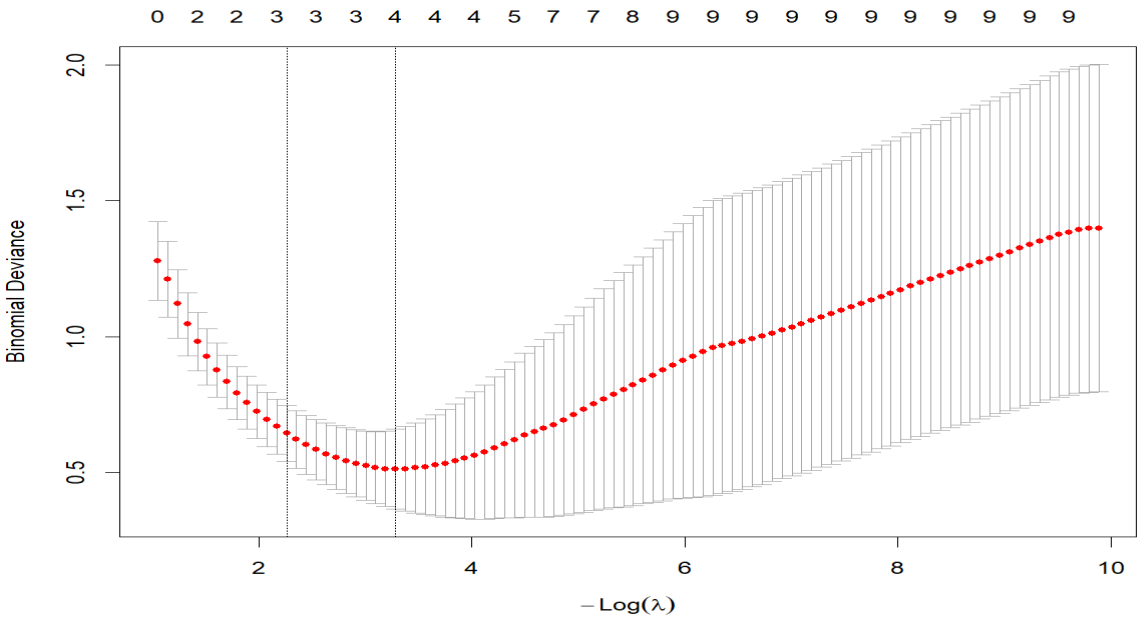

Fig:12 LASSO regression

Model Interpretability Using SHAP Analysis

The importance of features plot showed that a few genes carry out an exceptional weight on the accuracy of predictions, implying their high biological significance. The SHAP value plots per sample confirmed the effect of gene expression values in classifying samples.

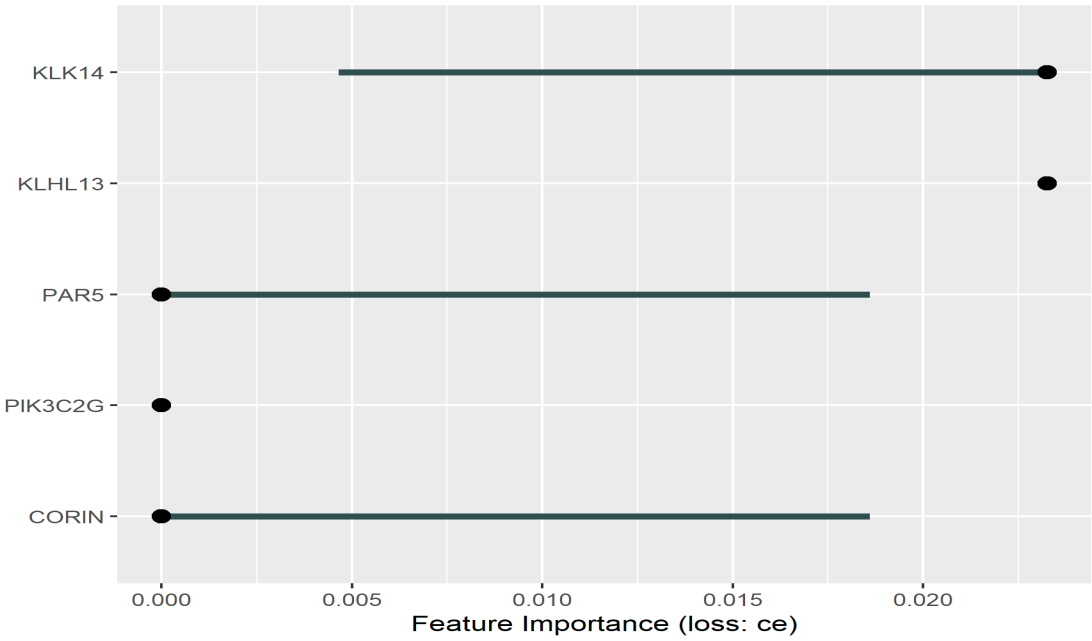

Fig:13 SHAP feature importance

In order to increase the interpretability of the machine learning model, SHAP analysis was conducted. The results of SHAP analysis provided information about the contribution of individual genes in the predictions of the model.

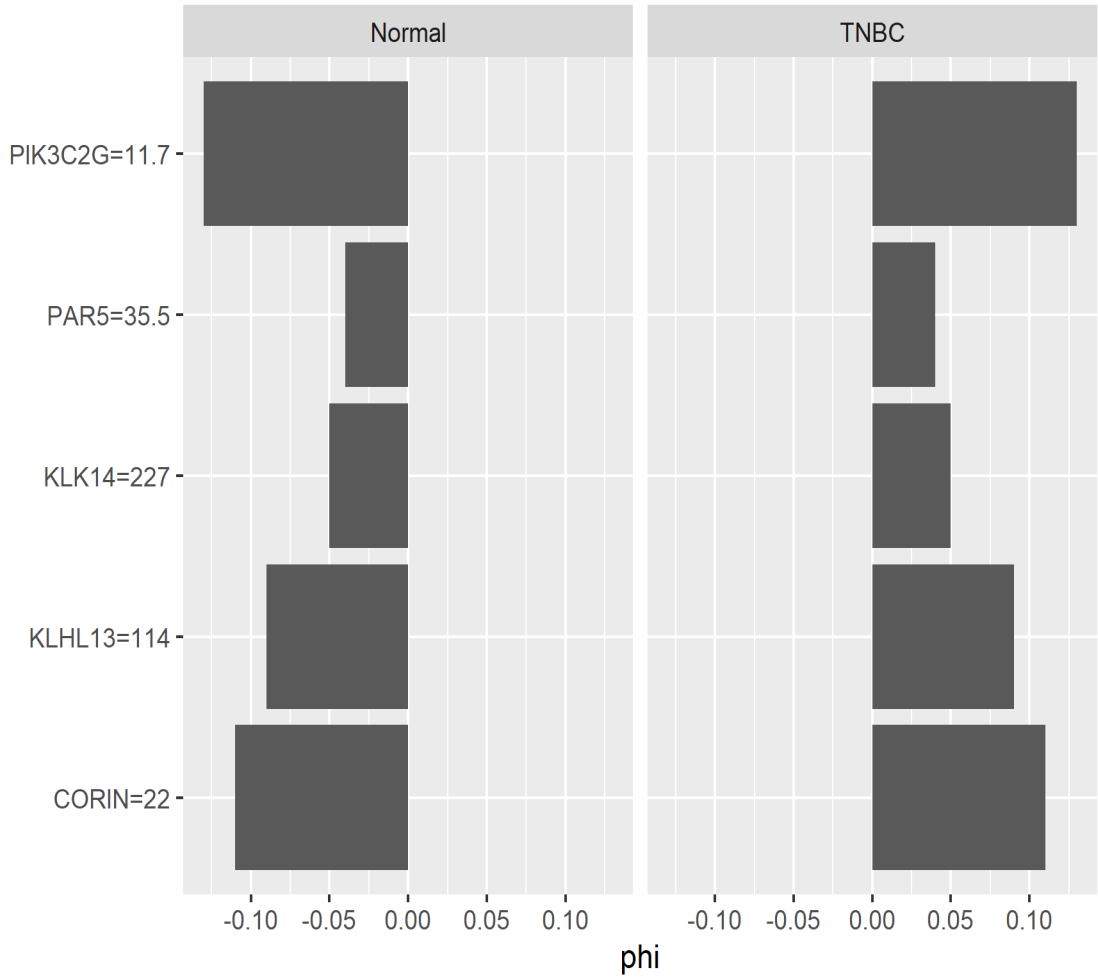

Fig:14 SHAP feature importance in Normal and TNBC

The graph illustrates the contribution of each individual gene to the classification of the particular sample. It demonstrates the influence of each gene on the outcome. Such an interpretive approach is necessary in clinical practice, as it helps build trust in the results generated by the model.

Discussion

Triple-negative breast cancer (TNBC) is a subtype of breast cancer The current study used an integrative analysis and machine learning approach based on dataset GSE38959 to find key biomarkers and potential therapy targets in TNBC. Results of the differential expression analysis showed significant transcriptional alterations among TNBC and normal samples, confirmed by evident differences between two types in PCA and heatmap analysis. Functional analysis suggested that differently expressed genes were related to the cell cycle process, DNA replication and mitosis, indicating a highly proliferative activity in TNBC. ROC analysis allowed us to find several genes with high discriminative power such as KLHL13, CORIN, PIK3C2G, KLK14, and STMN1, and the possibility of these genes to become biomarkers to distinguish TNBC from normal tissue was discussed. Genes are involved in signal transduction, proteolytic activity, and cytoskeleton regulation, being responsible for tumour progression and metastases. Analysis of the protein–protein interaction network made it possible to define some hub genes, e.g., CDC2 (CDK1), HSP90AA1, ASPM, MELK, and CENPK, which play an important role in cell cycle and mitosis, indicating a highly active proliferation of TNBC tumours. One of the reasons leading to the uncontrolled growth of tumours is over-expression of such genes that may also indicate a raised level of proliferation of cancerous cells. Among the genes that significantly interact with drugs are the metabolically active CYP1B1 NQO1 AKR1C1, and CYP1A1 ones, which are involved in xenobiotic metabolism and oxidative stress pathways, among other things. In particular, the CYP1B1 gene has been demonstrated to interact with anti-cancer drugs such as paclitaxel and docetaxel, whereas NQO1 gene was found to be associated with drug doxorubicin and mitomycin-C, thus suggesting the possibility of drug repositioning. The significance of the selected genes was confirmed through various machine learning algorithms including Random Forest and LASSO Regression methods, while SHAP analysis could help interpret the results by showing the most significant predictive features. In summary, several genes associated with diagnosis, hubs, and drugs have been implicated in the pathogenesis of TNBC, further studies will be needed for validation.

Summary

This study describes the implementation of an exhaustive bioinformatics pipeline to screen for biomarker candidates and drug targets associated with triple-negative breast cancer (TNBC) through the analysis of gene expression profiles. The data used for the research were sourced from the Gene Expression Omnibus (GEO). A differential gene expression test using the limma package yielded a substantial number of genes that are both up- and down-regulated in TNBC compared to normal breast tissue samples. The results obtained were visualized using tools like Volcano plot, Principal Component Analysis (PCA), and heat maps showing clear clustering of cancerous samples away from the normal samples.

Gene function enrichment studies, using GO and KEGG pathway analysis, showed that the screened genes are enriched in cellular proliferation processes such as mitosis, cell cycle control, and DNA replication. GSEA further confirmed the enrichment of the above functions among the identified genes. Machine learning techniques, including the application of Random Forest and LASSO regression models, were performed on the dataset to determine predictive biomarkers. ROC curve analyses showed that some of the screened genes were characterized by high diagnostic abilities, indicated by AUC values The PPI network analysis found some important hub genes like CYP1B1, NQO1, and AKR1C1 that have major involvement in tumour development and metabolism. The drug-gene interactions showed some promising drugs that could be considered for repositioning purposes. These include paclitaxel, doxorubicin, and tamoxifen, which interact with target genes. The SHAP analysis helped understand the contribution of individual genes to the model's predictions.

Conclusion

The current study is able to combine differential expression analysis, functional annotation, machine learning, and network biology methods to discover novel molecular biomarkers related to TNBC. Genes, especially those having good ROC performance metrics and central nodes in PPI networks, can serve as potential biomarkers for both diagnosis and prognosis. Participation of the genes in biological processes, such as cell cycle regulation and DNA replication, shows how important these molecules are in terms of cancer development.

At the same time, the drug–gene interaction analysis adds additional value to the research and helps discover new compounds capable of targeting specific oncogenes. Another useful technique used throughout the paper is the SHAP analysis which makes the models more interpretable and applicable in practice.

In conclusion, it should be noted that the proposed methodological strategy for the discovery of biomarkers is a rather powerful tool that can be used in both scientific and clinical environments. It is advisable to conduct future experiments using experimental and clinical databases, including TCGA.

Scope for further enhancement

The current analysis could be reinforced by the incorporation of other datasets such as The Cancer Genome Atlas to examine the reproducibility of the obtained results. Multi-cohort analysis will allow for a larger sample size which will enhance the external validity of the study. In future, the importance of identified key genes in relation to patient survival could be analysed via survival-based methods including Kaplan–Meier curves and Cox regression analysis. Development of nomograms based on gene expression combined with clinical features is likely to facilitate patient stratification in TNBC patients. Another aspect that could strengthen the current analysis includes incorporation of multi-omics data, including proteomics and epigenomics, for the sake of gaining further insight into molecular mechanisms driving the disease progression. Single-cell sequencing may prove useful in investigating heterogeneity of the tumour cells in addition to revealing biomarkers at the cellular level. Employing machine learning and deep learning algorithms could further advance the current findings in terms of accuracy and feature selection. Experimental validation of the current computational analysis using qRT-PCR, Western blotting, and immunohistochemistry is necessary. Furthermore, interactions between drugs and genes involved in tumour development could be confirmed via docking simulations and in vitro drug sensitivity analysis.

Last but not least, the creation of a web-based diagnostic or prognostic tool, as well as the use of AI approaches in large datasets, may assist in implementing the predictive models clinically.

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